Does Physical Activity Buffer the Association Between Screen Time and Insufficient Sleep Among U.S. High School Students?

Evidence from the Youth Risk Behavior Survey, 2015–2023 · Summer Research Camp (4 Weeks, Grades 10–11)

1. Background & Gap

The link between screen time and poor sleep in adolescents is established (r ≈ 0.25; five meta-analyses, 40+ studies). Only 23% of U.S. high school students get ≥8 hours of sleep (CDC, 2023). The open question is what *moderates* this association.

In October 2024, CDC published its first analysis of 2023 YRBS social media data (Young et al., MMWR 2024), finding frequent social media use was associated with sadness/hopelessness and suicide ideation. The authors explicitly identified what they did *not* examine:

> "Future research exploring the pathways through which social media use might lead to poor mental health and suicide risk… also is needed." — *Young et al., MMWR Suppl. 2024;73(4):1–10*

Notably absent from their analysis: **sleep as a pathway** and **physical activity as a protective factor**.

Why moderation, not mediation?

Dai & Ouyang (2026) showed that PA **mediates** the screen time ⟶ mental health pathway (screen time reduces PA, which harms mental health). We ask the complementary question: **does being physically active *protect* you from the sleep-disrupting effects of screen time, even if screen time stays high?**

This distinction matters for intervention: mediation says "reduce screen time to increase PA"; moderation says "increase PA to buffer against screen time you can't or won't reduce."

Hoe et al. (2019) tested this with NHANES accelerometry data (N=542, ages 16–19). Males meeting both PA and screen time guidelines had 73% lower odds of poor sleep *quality* (OR=0.27), but PA was associated with *less* sufficient sleep *duration* (OR=0.50) — possibly due to early-morning practices. No study has tested this moderation in U.S. high school students across multiple YRBS waves.

2. Research Questions

#	Question	Hypothesis
RQ1	How has the co-prevalence of high screen time (≥4 hrs/day) and insufficient sleep (<8 hrs) changed among U.S. high school students, 2015–2021?	Both have increased; co-prevalence has grown, with a COVID-era spike in 2021.
RQ2	Does PA moderate the association between screen time and insufficient sleep? (Pooled 2015–2021)	Among students meeting PA guidelines (≥5 days/week), the screen time–sleep association is weaker. The buffering effect may differ by sex.
RQ3	(2023 only) Does frequent social media use predict insufficient sleep, and does PA moderate this?	Frequent SM use predicts insufficient sleep; PA moderates this association.

Design: Two-part study. RQ1–RQ2 use the screen time question (2015–2021). RQ3 uses the new 2023 SM frequency question. See Appendix A for variable details.

3. Conceptual Model

```
                      ┌───────────────────┐
                      │ Physical Activity  │
                      │    (Moderator)     │
                      │  0–1 / 2–4 / 5–7  │
                      │    days/week       │
                      └───────────────────┘
                               │
                       Does PA weaken
                       this association?
                               │
                               ▼
┌───────────────┐                            ┌─────────────────┐
│  Screen Time   │        association         │  Insufficient    │
│  (IV)          │ ─────────────────────>  │  Sleep <8 hrs     │
│  <2 / 2–3 / 4+│   (cross-sectional;       │  (DV)            │
│  hrs/day       │     possibly bidirectional) │                   │
└───────────────┘                            └─────────────────┘

Covariates: Sex, Race/Ethnicity, Grade, Sexual Identity,
            Cyberbullying, BMI (sensitivity), Survey Year
Unmeasured: School start times, SES, Caffeine, Chronotype

RQ3 (2023 only): Replace Screen Time IV with
SM Frequency (≥ several times/day vs. less)
```

No mediation arrows. Depression is reported as descriptive context in RQ1 trends only.

4. Data Source

CDC Youth Risk Behavior Survey (YRBS), 2015–2023 — nationally representative, ~13K–17K students per wave, freely downloadable. No IRB required (public, de-identified federal data).

Waves	Purpose	N (approx)
2015, 2017, 2019, 2021	RQ1–RQ2 (screen time)	~56,000 pooled
2023	RQ3 (SM frequency)	~17,000

⚠️ **Day 1 blocking task:** The general screen time question was likely dropped in 2023 (replaced by SM frequency). Download raw 2023 data and verify. If absent, RQ1–RQ2 use 2015–2021 only; RQ3 uses 2023 SM frequency as planned.

See Appendix A for key variables, covariates, and known data limitations.

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5. Analysis Plan

All analyses use R's `survey` package for complex survey design (weights, strata, PSUs). Pooled weights = original weight ÷ number of years combined (per CDC guidance).

RQ	Method	Key Output
RQ1	Weighted prevalence of high screen time, insufficient sleep, and co-occurrence by year × sex. Log-linear model testing whether co-prevalence grew beyond marginal trends.	Figure 1 (trend lines) + Table 1
RQ2	Weighted logistic regression: insufficient sleep ~ screen time × PA + covariates (pooled 2015–2021). Stratified AORs by PA level if interaction is significant. Three-way interaction (× sex) is exploratory.	Tables 2–3 + Figure 2 (interaction plot)
RQ3	Weighted logistic regression: insufficient sleep ~ SM frequency × PA + covariates (2023 only).	Table 4 + Figure 3

Sensitivity analyses: (1) Exclude 2021 (COVID break), (2) Sleep as ordinal vs. binary, (3) Restrict to identical-wording years, (4) Models with/without BMI (potential collider). See Appendix B for details.

Pre-registration: Register on OSF *before* downloading data. See Appendix F for template.

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6. Four-Week Timeline

Week	Focus	Key Milestones
1	Foundation	Pre-register on OSF (Mon). Download data, **verify screen time variable availability** (Tue — blocking). Clean data, compare question wording across years, literature review.
2	Analysis	RQ1 trends (Mon). RQ2 moderation + sensitivity analyses (Tue–Wed). RQ3 SM frequency (Thu). Mid-project review with mentor (Fri).
3	Writing	Methods (Mon), Results (Tue), Introduction (Wed), Discussion (Thu), Assemble draft + Abstract (Fri).
4	Revision & Submit	Peer review (Mon), Revise (Tue), Format for journal (Wed), Final proofread (Thu), **Submit + upload preprint to OSF** (Fri 🎉).

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7. Team Roles (~4 students + mentor)

Role	Focus	Writes
Data Lead	Download raw CDC data, verify variables, clean, maintain R scripts/GitHub	Methods — Data Source
Analysis Lead	Run regressions, survey design, sensitivity analyses	Methods — Stats + Results
Literature Lead	Coordinate reading, Zotero references, annotated bibliography	Introduction + Discussion
Visualization Lead	Figures, manuscript formatting, supplementary tables	Figures/Tables + Abstract
Mentor	Verify survey weights Day 1, review stats Week 2, review draft Week 4	Final review + cover letter

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8. Expected Contributions

Scientific

- Tests PA as **moderator** (buffer) vs. Dai & Ouyang's mediator finding
- Extends Hoe et al. (2019) from NHANES (N=542, one cross-section) to YRBS (N~56K, 4 waves)
- Responds directly to CDC's MMWR 2024 gap statement
- Early analysis of 2023 YRBS SM-specific items in the context of sleep and PA

Practical

- **If PA buffers:** Schools can promote PA as a concrete response — "exercise more" is more actionable than "use your phone less"
- **If PA doesn't buffer:** Challenges the assumption that exercise compensates for screen time — implies screen time reduction is necessary
- **If sex differences exist:** Interventions may need tailoring (Hoe et al. found PA buffered for boys but not girls on sleep quality)

For Students

Publishable research at scale, pre-registration and open science, R programming, survey data analysis, academic writing — foundation for college applications and research careers.

Appendix A: Variables & Data Details

Key Variables

Construct	YRBS Item	Operationalization
Screen time (IV) — RQ1–RQ2	"On an average school day, how many hours do you play video or computer games or use a computer for something that is not school work?"	Low (<2 hrs), Moderate (2–3 hrs), High (≥4 hrs)
Sleep duration (DV)	"On an average school night, how many hours of sleep do you get?"	Binary: Insufficient (<8 hrs) vs. Sufficient (≥8 hrs), per AASM ages 13–18. Also ordinal in sensitivity analysis.
Physical activity (Moderator)	"During the past 7 days, on how many days were you physically active for a total of at least 60 minutes per day?"	Low (0–1 day), Moderate (2–4 days), High (5–7 days)
SM frequency (IV) — RQ3 only	"How often do you use social media?"	Binary: Frequent (≥ several times/day) vs. Infrequent, per CDC MMWR dichotomization
Depressive symptoms	"...did you ever feel so sad or hopeless almost every day for two weeks or more...?"	Binary. Descriptive context in RQ1 only; not modeled as outcome.

Covariates

Covariate	Justification
Sex	Females report worse sleep and more SM use
Race/ethnicity	Black students report less sleep; disparities documented
Grade	Older students report less sleep
Sexual identity	LGBTQ+ youth have worse sleep/mental health; harmonize coding across waves
Cyberbullying	CDC's 2023 analysis identified bullying as key pathway
BMI	Associated with both screen time and sleep; **potential collider** — run with and without
Survey year	Controls for secular trends when pooling

Unmeasured Confounders (Limitations)

Confounder	Why It Matters
School start times	Strongest structural determinant of adolescent sleep duration
SES	Associated with screen time patterns and sleep environment; no income/education variable in YRBS
Caffeine	Associated with evening screen use and sleep disruption
Chronotype	Biological variation confounds both PA and screen time associations

Screen Time Question Availability

Years	Status
2015, 2017, 2019	✅ Available (wording may vary in platform examples)
2021	⚠️ Verify — may have combined TV + digital device use
2023	❌ Likely dropped; replaced by SM frequency question

Known Measure Limitations

Limitation	Mitigation
Screen time bundles SM with gaming (pre-2023)	RQ3 uses 2023 SM-specific question
SM frequency is a single crude item	Follow CDC's validated dichotomization; note as future research direction
Question wording evolved across years	Document exact wording (Appendix D); sensitivity analysis on identical-wording years
2023 shifted from paper to tablet-based survey	CDC studies found mode did not affect prevalence estimates
Self-report bias	Inherent to all YRBS research; good test-retest reliability documented
Reverse causation (can't sleep → pick up phone)	Cannot resolve with cross-sectional data; discuss prominently

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Appendix B: Sensitivity Analyses & Statistical Notes

Sensitivity Analyses (RQ2)

1. **Exclude 2021** (COVID structural break — fall vs. spring administration)
2. **Sleep as ordinal** (hours) rather than binary — ordinal/linear regression with survey weights
3. **Restrict to identical-wording years** (likely 2015–2019)
4. **Models with and without BMI** — BMI may be a collider (screen time ⟶ reduced PA ⟶ higher BMI ⟶ worse sleep); report both, note discrepancies

Three-Way Interaction Protocol

Before running Screen time × PA × Sex: tabulate cell sizes. If any cell has unweighted N < 50, collapse categories (e.g., binary PA). Three-way results are explicitly exploratory.

Multiple Comparisons

No formal corrections (e.g., Bonferroni) applied to exploratory interaction tests, following conventions in observational epidemiology. Borderline interactions ($0.01 < p < 0.05$) interpreted with caution; emphasis on effect sizes and confidence intervals.

Statistical Power

Pooled N~56K (RQ2) and ~17K (RQ3) provide strong power for main effects. Interaction effects require 4–16× sample size of main effects (Aguinis, 1995). Sample is adequate for two-way interactions of moderate effect size but may be underpowered for the three-way interaction.

Appendix C: Risks & Mitigations

Risk	Likelihood	Impact	Mitigation
Screen time question absent from 2023	High	High	Blocking Day 1 verification. If absent: RQ1–2 use 2015–2021; RQ3 uses 2023 SM frequency.
Screen time wording changed in 2021	Medium	High	Download questionnaire PDF Day 1. If changed: sensitivity analysis excluding 2021.
PA × screen time interaction is null	Medium	Medium	Null is publishable: "PA does not buffer" is informative and challenges assumptions.
PA associated with *worse* sleep duration	Medium	Medium	Consistent with Hoe et al. (early-morning practices). If replicated, discuss mechanisms.
COVID structural break biases 2021	High	Medium	Flag 2021 on charts. Report models with and without 2021.
BMI as collider	Medium	Medium	Report models with and without BMI.
YRBS data access instability	Medium	Medium	Download and archive all raw data locally Day 1. `dissertationData` R package as backup.
`dissertationData` lacks survey design vars	Medium	High	Use raw CDC data as primary source; package is backup only.
Reverse causation	High	Medium	Discuss prominently in limitations. Cannot resolve with cross-sectional data.
Competitor publishes similar analysis	Low	High	Pre-register on OSF to timestamp priority. Our specific combination is unique.
Three-way interaction cell sizes too small	Medium	Medium	Tabulate before modeling. Collapse PA to binary or report two-way stratified by sex.
R learning curve for high schoolers	Medium	Medium	Mentor provides starter code. Analysis Lead works in R; others focus on writing/visualization.
Manuscript rejected by first journal	Medium	Low	Preprint on OSF for immediate visibility. Submit to backup journals.

Appendix D: Question Wording Comparability Checklist

Download questionnaire PDFs from CDC and complete this table. **Blocking task — results determine the study design.**

Year	Screen Time Question Text (exact)	Example Platforms	Response Options	Survey Mode	Identical to 2019?	Variable Present?
2015				Paper		✅ Verify
2017				Paper		✅ Verify
2019				Paper	Baseline	✅ Verify
2021				Paper (fall)		⚠️ May differ
2023				Tablet		❌ Likely absent

Also check: CDC YRBS Questionnaire Content 1991–2023 document. Include completed table as Supplementary Table S1.

Appendix E: Publication Venues

Journal	Fit	Review	APC	Notes
Cureus	✅ Best	2–4 wks	Free	Open access; publishes student work; PubMed indexed
Journal of Student Research	✅ Excellent	4–6 wks	Free	For student-authored research
IJHSR	✅ Good	4–8 wks	Free	For high school researchers
PLOS ONE	🟡 Stretch	8–12 wks	~$2K	Broad scope; rigorous methodology required
Frontiers in Public Health	🟡 Stretch	8–12 wks	~$3K	Open access; accepts observational studies

Strategy: Submit to Cureus first. Upload preprint to OSF simultaneously. If rejected, revise for JSR or IJHSR.

Appendix F: Pre-Registration Template (OSF)

Submit Day 1, Monday — BEFORE downloading data.

```
Title: Does Physical Activity Buffer the Association Between Screen Time
       and Insufficient Sleep Among U.S. High School Students?

Authors: [Student names], [Mentor name]

Data source: CDC YRBS 2015–2023 (public, de-identified)
  - RQ1–RQ2: 2015–2021 pooled (screen time variable)
  - RQ3: 2023 only (social media frequency variable)
  Note: If Day 1 verification reveals screen time variable exists in
  2023, RQ1–RQ2 will expand to 2015–2023.

Research Questions:
  RQ1: Trend in co-prevalence of high screen time and insufficient
       sleep (2015–2021), with formal test of co-prevalence growth
  RQ2: PA × screen time interaction on insufficient sleep (pooled
       2015–2021)
  RQ3: (2023) SM frequency predicts insufficient sleep; PA moderates

Primary outcome: Insufficient sleep (<8 hrs on school nights), binary
Primary predictor: Screen time (categorical: <2, 2–3, ≥4 hrs/day)
  for RQ1–RQ2; SM frequency (binary) for RQ3
Moderator: Physical activity (categorical: 0–1, 2–4, 5–7 days/week)
Covariates: Sex, race/ethnicity, grade, sexual identity, cyberbullying,
            BMI, survey year (RQ1–RQ2 only)

Analysis: Weighted logistic regression with interaction term using
          survey package in R. Stratified AORs if interaction p < 0.05.
          Three-way interaction (× sex) is exploratory.

Sensitivity analyses:
  1. Exclude 2021 (COVID structural break)
  2. Sleep as ordinal (hours) vs. binary
  3. Restrict to years with identical question wording
  4. Models with and without BMI (collider check)

Limitations acknowledged a priori:
  - Cross-sectional design; reverse causation possible
  - Unmeasured confounders: school start times, SES, caffeine, chronotype
  - Multiple models without formal correction; interpret borderline
    interactions with caution
  - Hoe et al. (2019) found PA may reduce sleep duration even as it
    improves quality; a null or adverse finding for duration is plausible

This study was pre-registered BEFORE any data were accessed.
```

Appendix G: Essential Reading List

Must-Read (assign across team)

#	Citation	Relevance
1	**Young et al. (2024)** — MMWR Suppl. 73(4):1–10. CDC's 2023 SM analysis.	Contains the gap statement we respond to. Must cite and quote.
2	**Dai & Ouyang (2026)** — *Humanities & Social Sciences Communications* 13, 256.	Most direct competitor. PA as mediator (NSCH, ages 6–17). We test moderator (YRBS, high school).
3	**Hoe et al. (2019)** — *Int J Environ Res Public Health* 16, 1524.	PA × screen time on sleep quality (not duration) via NHANES (N=542). We extend with YRBS (N~56K).
4	**CDC YRBS Data Summary & Trends (2025)** — Dietary, PA, Sleep: 2013–2023.	Baseline trend data. Only 23% meet sleep guidelines.
5	**PMC12652926 (2024)** — Excessive Screen Time Among U.S. High School Students.	Similar methodology (YRBS 2019). Documents 2023 dropped general screen time question.

Methods References

#	Citation	Purpose
6	CDC "Combining YRBS Data Across Years and Sites"	Required for pooling survey weights
7	CDC "Software for Analysis of YRBS Data"	Survey weight handling in R
8	CDC 2023 YRBS Data User's Guide	Codebook, variable definitions
9	CDC YRBS Questionnaire Content 1991–2023 (PDF)	Which questions in which years

Background (skim)

#	Citation	Purpose
10	Chen et al. (2024) — Meta-analysis, 40 studies	SM ⟶ sleep landscape
11	Khan et al. (2024) — HBSC, 212K, 40 countries	SM ⟶ sleep in adolescents
12	Scoping review of reviews (2025) — PMC12840076	11 reviews reviewed
13	Maxwell & Cole (2007) — Bias in cross-sectional mediation	Why we dropped mediation

Appendix H: Starter R Code

```r
# ============================================================
# STEP 1: Download raw CDC data (PRIMARY SOURCE)
# ============================================================
# https://www.cdc.gov/yrbs/data/national-yrbs-datasets-documentation.html
library(survey)
data(yrbs)  # small example bundled with survey package
names(yrbs) # → "weight" "stratum" "psu" "qn8" ...

# ============================================================
# STEP 2: BLOCKING VERIFICATION (Day 1)
# ============================================================
# After loading each year's data:
# (a) Confirm survey design variables exist
survey_vars <- c("weight", "stratum", "psu")
# all(survey_vars %in% tolower(names(yrbs_2023)))

# (b) Check if screen time variable exists in 2023
# Look for Q79, Q80, Q81 or similar
# If NOT found → confirm two-part study design

# (c) Check 2021 screen time question wording vs. 2019

# ============================================================
# STEP 3: SURVEY DESIGN SETUP
# ============================================================
library(survey); library(dplyr); library(ggplot2)
options(survey.lonely.psu = "adjust")

n_years <- 4  # 2015, 2017, 2019, 2021
yrbs_pooled <- yrbs_combined %>%
  filter(year %in% c(2015, 2017, 2019, 2021)) %>%
  mutate(pooled_weight = weight / n_years)

yrbs_design <- svydesign(
  ids = ~psu, strata = ~stratum,
  weights = ~pooled_weight, data = yrbs_pooled, nest = TRUE
)

# ============================================================
# RQ1: DESCRIPTIVE TRENDS
# ============================================================
yrbs_design <- update(yrbs_design,
  insuff_sleep = as.numeric(sleep_hours < 8),
  high_screen  = as.numeric(screen_time_hrs >= 4)
```

```
)
trend_sleep <- svyby(~insuff_sleep, ~year, yrbs_design, svymean, na.rm=TRUE)

# ============================================================
# RQ2: MODERATION (PA × Screen Time → Sleep)
# ============================================================
yrbs_design <- update(yrbs_design,
  pa_cat = cut(pa_days, breaks = c(-1, 1, 4, 7),
               labels = c("Low (0-1)", "Moderate (2-4)", "High (5-7)"))
)

model_mod <- svyglm(
  insuff_sleep ~ screen_cat * pa_cat + sex + race_eth + grade +
    sexual_identity + cyberbullying + bmi_cat + factor(year),
  design = yrbs_design, family = quasibinomial()
)

# Sensitivity: without BMI (collider check)
model_no_bmi <- svyglm(
  insuff_sleep ~ screen_cat * pa_cat + sex + race_eth + grade +
    sexual_identity + cyberbullying + factor(year),
  design = yrbs_design, family = quasibinomial()
)

# Test interaction significance
library(car)
linearHypothesis(model_mod, grep("screen_cat.*pa_cat",
  names(coef(model_mod)), value = TRUE))

# Stratified AORs by PA level
for (pa in levels(yrbs_design$variables$pa_cat)) {
  sub <- subset(yrbs_design, pa_cat == pa)
  m <- svyglm(insuff_sleep ~ screen_cat + sex + race_eth + grade +
    sexual_identity + cyberbullying + bmi_cat + factor(year),
    design = sub, family = quasibinomial())
  cat("\n--- PA Level:", pa, "---\n")
  print(exp(cbind(AOR = coef(m), confint(m))))
}

# ============================================================
# RQ3: SM FREQUENCY (2023 only)
# ============================================================
yrbs_2023_design <- svydesign(
  ids = ~psu, strata = ~stratum, weights = ~weight,
  data = yrbs_2023, nest = TRUE
)
model_sm <- svyglm(
  insuff_sleep ~ sm_frequent * pa_cat + sex + race_eth + grade +
    sexual_identity + cyberbullying + bmi_cat,
  design = yrbs_2023_design, family = quasibinomial()
)
```

Appendix I: Ethical Considerations

- **No IRB required.** Secondary analysis of publicly available, de-identified federal data.
- **No identifiable data.** No names, school identifiers, or sub-state geographic identifiers.
- **Responsible reporting.** Associations framed as associations, never causation. Reverse causation acknowledged. Avoid deficit framing of racial/ethnic groups. Null findings reported honestly.
- **Pre-registration.** Analysis plan registered before data access to prevent p-hacking.